Monte Carlo Simulations to Model Epidemic Spreading

Chelsea O'Sullivan, Matthew Amoo, Kevin O'Reilly

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1 Author contributions

Chelsea wrote Chapters 3.1, 3.2, 3.4, 4.2, and 6.

Matthew wrote Chapters 3.3, 4.1 and 5.

Kevin wrote Chapters 2, 3.5, 4.3 and 7.

2 Introduction

Understanding how diseases spread throughout a population is of massive scientific and medical importance. Traditional epidemiological models often assume homogenous mixing amongst its population. In reality, populations are structured, and people interact through non-uniform patterns of contact. Network science can provide a framework to capture this heterogeneity, allowing us to model how infections propagate through realistic social systems. A network simply consists of nodes and edges which connects certain nodes to one another. In our modelling of epidemic spreading, nodes represent people, and edges connecting people to each other represent their contacts. In this project, three network models, namely Erdős-Rényi random graphs, Watts-Strogatz small world networks and the Barabási–Albert scale-free network, are implemented to analyse infection spreading. We use random walks to model diffusion across the networks. In a random walk, we begin at a given node and randomly move to a neighbouring (i.e. connected) node iteratively. Monte-Carlo simulation is used to undergo repeated random walks of the network to generate, via the law of averages, estimates of average case scenarios of how the infection spreads, looking at outcomes such as outbreak size or time to peak infection rate. At each time step in a random walk or epidemic process, probabilistic events such "infect this neighbour with probability β", or "recover with probability γ" occur. Rather than attempting to solve the full system analytically, we simulate these events many times over a given network. Each simulation represents an encountered realisation of the epidemic. Through analysing our collection of outcomes, we can produce predictions and generate probabilities associated with certain occurrences.

3 Algorithm Design

3.1 SIR Models for Epidemic Spreading

SIR

In epidemiology, the standard model for mapping epidemic spreading is some variation of an SIR model. SIR models categorise a population into three groups: susceptible (S), infected (I), and recovered (R). These models use two variables to map changes from S to I, and I to R. These are: β (the transmission rate) and γ (the recovery rate). γ is inversely proportional to the infectious period of a disease (gamma = 1/infectious period). This means that diseases that have long infectious periods (HIV, TB) will have small γ values, and diseases with short infectious periods (flu, Covid19) will have large γ values. β also will vary depending on the

disease being modelled, for example airborne diseases like measles will have high beta, while diseases requiring physical contact like HIV will have lower β. Together, these parameters and the three SIR states can be expressed as a set of differential equations:

$$\frac{dS}{dt} = -\beta \tfrac{SI}{N}, \ \frac{dI}{dt} \ = \beta \tfrac{SI}{N} - \gamma I, \ \frac{dR}{dt} = \gamma I. \tag{1}$$

where β is the transmission rate and γ is the recovery rate.

SEIR

A common variation to the standard SIR model is the SEIR model, which incorporates an extra category, those who have been exposed but are not yet infected (E). This model is more true to real-life than SIR, as most infectious diseases have an incubation period. This model introduces an extra variable, σ, which represents the rate at which those who are exposed transition to being infected. σ is inversely proportional to the incubation period, so diseases that have long incubation periods (i.e. TB) will have small sigma values. As differential equations the SEIR model can be expressed as:

$$\frac{dS}{dt} = -\frac{SI}{N}\beta, \ \frac{dE}{dt} \ = \frac{SI}{N}\beta - \sigma E, \ \frac{dI}{dt} = \sigma E - \gamma I, \ \frac{dR}{dt} \ = \gamma I, \tag{2}$$

where β is the transmission rate, σ is the rate at which exposed individuals become infectious, and γ is the recovery rate.

SEIRD

SEIRD builds on SEIR, but includes a final category: those who have died (D). This model is especially important for diseases that have high mortality rates (Ebola, Marburg). This model has the parameter μ which represents the probability each person has of dying per unit time. The SEIRD model can also be expressed as differential equations:

$$\frac{dS}{dt} = -\frac{SI}{N}\beta, \ \frac{dE}{dt} \ = \frac{SI}{N}\beta - \sigma E, \ \frac{dI}{dt} = \sigma E - (\gamma + \mu)I, \ \frac{dR}{dt} \ = \gamma I, \ \frac{dD}{dt} = \mu I, \tag{3}$$

where β is the transmission rate, σ is the rate at which exposed individuals become infectious, γ is the recovery rate, and μ is the disease-induced death rate.

3.2 Monte Carlo Simulations for SIR Models

The earlier differential equations used to express SIR models are deterministic models which are not the most appropriate model of epidemic spreading for a variety of reasons. These differential equations will allow for fractions, i.e. 98.3 infected people, which doesn't make sense for a real-life context. These models also don't accurately model social interactions. They assume that every person has an equal probability of meeting any other person, and don't allow for stochastic effects such as superspreaders. Deterministic models also don't allow for the possibility that a potential epidemic could die out before it ever reaches epidemic status i.e. if 50 people are all infected but don't go on to infect anyone else.

Monte Carlo simulations don't have these issues so they can be used to more accurately model epidemic spreading. While deterministic PDEs use continuous SIR variables to compute mean-field approximations, network-based Monte Carlo simulations simulate random discrete interactions based on more sophisticated contact structures. An important aspect in network-based Monte Carlo simulations is choosing the network. There are many options which seek to model different aspects of real like interactions, i.e. Watts-Strogatz which models clustered small-world structures, and Barabási–Albert which incorporates superspreaders.

3.3 The Erdős-Rényi Network

The ER network is based on a random graph, $G(n, p)$. It is among the simplest network models for understanding random interaction patterns. It starts with n nodes and then, for every possible pair of nodes, an edge is added with a probability of p. The ER network is stochastic in nature, meaning that a different graph will be generated every time the model is run, even if you use the same n and p values.

A key property of ER networks is that they are homogeneous contact networks. This means that most nodes have similar numbers of neighbours, with degrees following a binomial distribution. When n is large and p is small, the degrees begin to follow a Poisson distribution. This means that infection spreads in a relatively uniform and unbiased way. The level of clustering is very small, so groups of neighbours in the network would resemble trees, which is helpful for the behaviour of random walks.

All of these things make ER networks a useful baseline or 'dummy' model for studying diffusion under uniform random mixing. In the context of epidemic spreading, ER networks allow epidemics to be illustrated without the additional structural effects present in more realistic networks, such as Watts-Strogatz or Barabási-Albert graphs.

3.4 The Watts-Strogatz Network

The Watts-Strogatz network starts with a regular ring lattice with a high amount of clustering. The network then rewires some edges to link to a different node further away, introducing a shortcut. This has the effect of dramatically reducing the average path length between nodes, while retaining a high level of clustering. In doing so, the Watts-Strogatz network combines two properties that are seen in real social interaction networks. High clustering (the idea that your friends are likely to know each other), and short random walks. Short random walks mean that it shouldn't take many steps for one person to connect to another through their acquaintances. This idea was coined in the 1980s by the American psychologist Stanley Milgram, who tasked a group of people in the mid-West with sending a letter to a stranger in Boston by passing it to people they knew on a first name basis, who would then pass it to someone else, and so on. On average it took six connections for the letter to reach the stranger in Boston. Milgram then came up with the concept of "six degrees of separation", the idea that we are all just 6 social connections away from each other. This is what the Watts-Strogatz network is modelling by reducing the average path lengths between nodes.

In practice, the Watts–Strogatz model works by choosing a rewiring probability p. Each edge in the original ring lattice is revisited, and with probability p it is "rewired" so that its endpoint is replaced by a randomly chosen node. When $p = 0$, the graph remains a perfectly regular lattice with long path lengths; when $p = 1$, the graph resembles a completely random network. Because edges are rewired to create shortcuts, random walks reach distant parts of the network far more quickly. This is one reason the model is used for studying diffusion processes, epidemic spreading, and information flow in social systems.

3.5 The Barabási–Albert Network

The Barabási–Albert algorithm is a model for generating undirected scale-free networks. Its key characteristic is the idea of preferential attachment. New nodes that join a network are more likely to be connected to a node which already has many connections. This further increases the well-connected node's links, thus increasing its probability to attach to future nodes. This self-fulfilling loop produces the "rich-get-richer" behaviour associated with this model, which is applicable to many real systems such as social networks. Most nodes have few edges, but a small number of hubs emerge. These hubs replicate superspreaders in our modelling of infection spreading. This describes the model's scale-free property, as the degree distribution of these networks follows a power-law, where there is no typical degree around

which all nodes cluster, instead the distribution is heavily tailed. Our Barabási–Albert models have two parameters. N is the total number of nodes in the completed network. The network begins with m nodes, all of degree 1. N-m nodes are then added one by one, and each is initialised with m connections. The total number of edges in the network is then equal to m(N-m). The probability of an existing node being one of the m to be connected to a new node entering is proportional to its degree. This process of adding nodes one by one can lead to earlier nodes generally ending up with higher degree, as if they happen to receive connections early it can generate a feedback loop.

4 Algorithm Demonstration

4.1 Erdős-Rényi Network

This section demonstrates how the SIR simulation behaves when applied to an ER network. The ER graph in this case is $G(100, 0.2)$, meaning there are 100 nodes and a 20% chance of an edge forming between two nodes.

A run begins with an ER network in which every node is susceptible. One at random is chosen to be infected, which marks the commencement of the outbreak. There are one hundred timesteps, and at each timestep, susceptible nodes become infected with probability beta if they have an infected neighbour. Infected nodes may recover with probability gamma. Recovered nodes are assumed to be immune for the rest of the simulation, following the Kendrick and McKendrick model [1]. At every timestep, the number of susceptible, infected, and recovered nodes is recorded to generate SIR curves for analysis.

As the simulation updates infection and recovery only once per timestep, the timing of events is essentially grouped into daily blocks. In real life, this isn't the case: some can become infected in the morning, and another in the afternoon. As a result, some infections that should happen earlier get pushed to the end of the day, and recoveries don't show up until the following timestep. This doesn't really change the overall shape of the outbreak curves, but it can nudge the timing of specific events in a single run and introduce some extra noise. Once you average many runs together, those little timing errors mostly cancel out, which is why the Monte Carlo results look much smoother (see end of Section 4).

To show the algorithm works under different conditions, three types of examples are provided: A single random walk, multiple examples with different infection probabilities and a Monte Carlo averaged simulation

4.1.1 Single Random Walk (SIR implementation)

The first example shows a single simulation of the SIR model on an Erdős–Rényi network with $\beta = 0.03$ and $\gamma = 0.01$ (Figure 1). Because infection and recovery are stochastic processes, each run produces a slightly different trajectory.

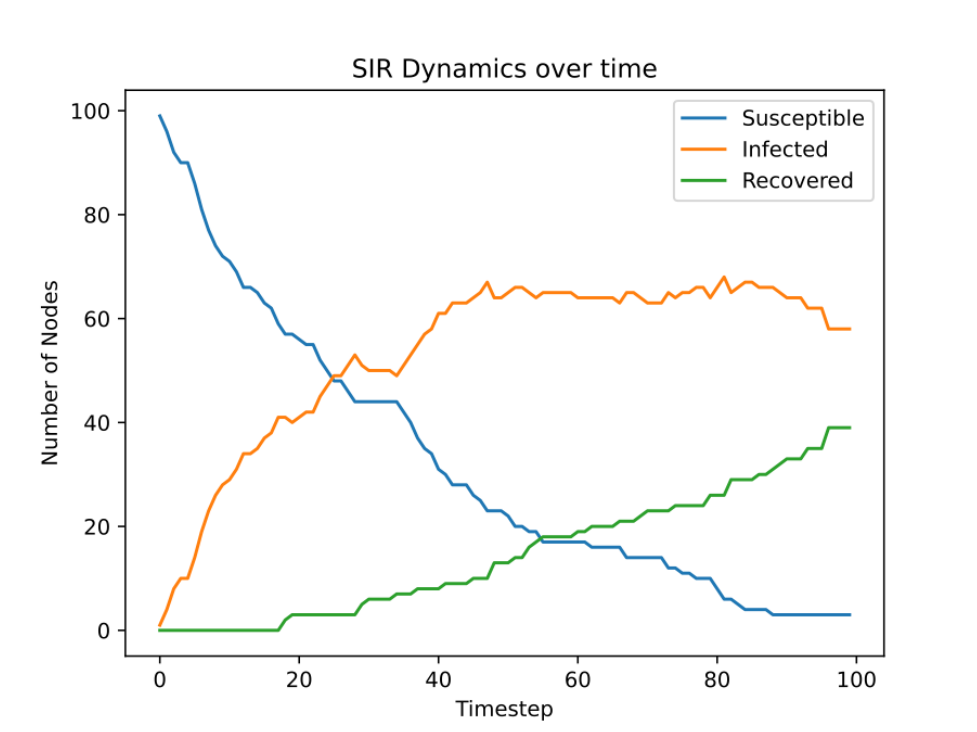

Figure 1: SIR Dynamics over time: Single Random Walk

Figure 1 shows that the number of infected individuals rises quickly within the first few timesteps. The infected curve peaks once most susceptible individuals become infected. It finally begins to decline as recovery becomes more likely than new infections.

4.1.2 Parameter Variation

To show how the algorithm behaves under different conditions, the model was run with different infection probabilities β while keeping the recovery probability fixed at $\gamma = 0.01$

Two contrasting examples are shown (Figure 2 and 3):

Lower Infection Probability ($\beta = 0.02$)

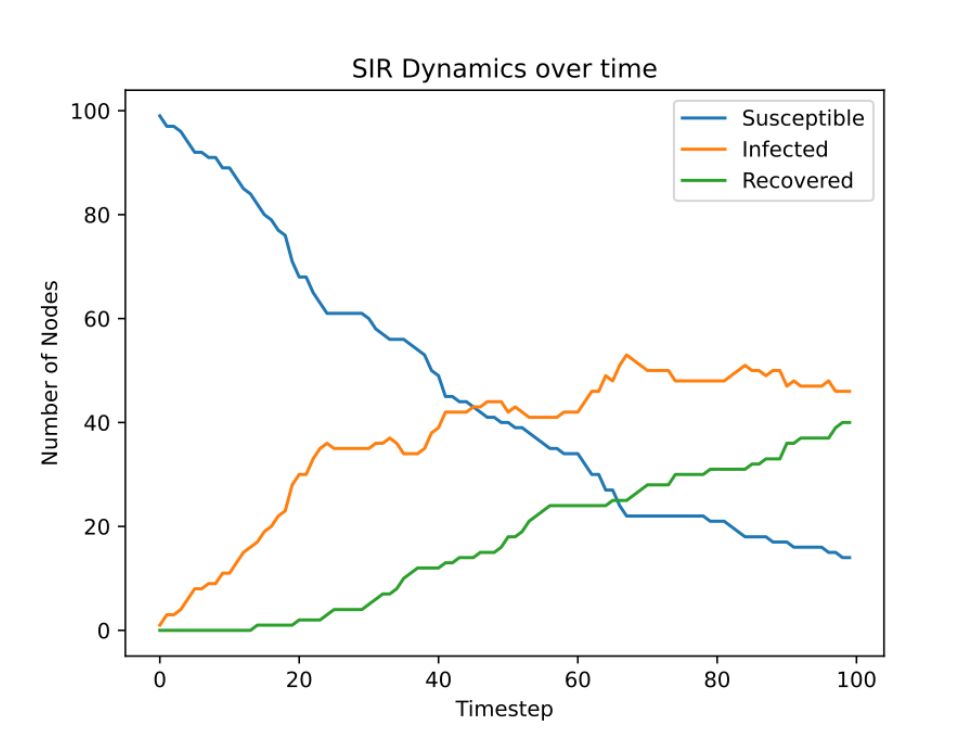

Figure 2: SIR Implementation (Lower β)

Figure 2 reveals that reducing infection probability results in a weaker epidemic outbreak. The infection population remains quite low (approximately less than 50%) in comparison to the plot from the first example.

Higher Infection Probability (β= 0.05)

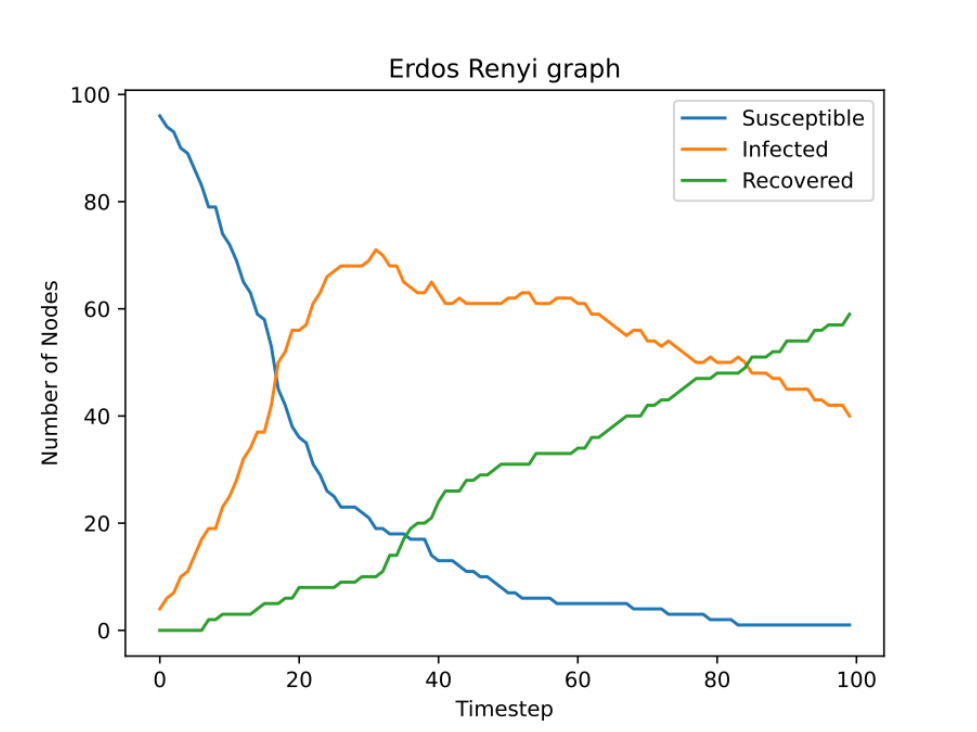

Figure 3: SIR implementation(Higher β

Figure 3 shows that increasing beta to 0.05 produces a much faster-growing outbreak. The infection curve rises sharply, peaks earlier, and infects a larger fraction of the network.

Changes to the recovery rate gamma aren't shown, but they would also result in changes to the epidemic curve. An increased recovery rate results in a lower infection period, and thus, the outbreak does not last long.

All three of these examples reveal the sensitivity of the epidemic curve to the rate of infection. However, the randomness in each run is clear to see, and this motivates the need for performing Monte Carlo simulations.

4.1.3 Monte Carlo Averaged Simulation

As single runs are noisy due to random infection and recovery events, the simulation was repeated 100 times, and the results were averaged at each timestep. Here, $\beta = 0.03$ and $\gamma = 0.01$ (Figure 4).

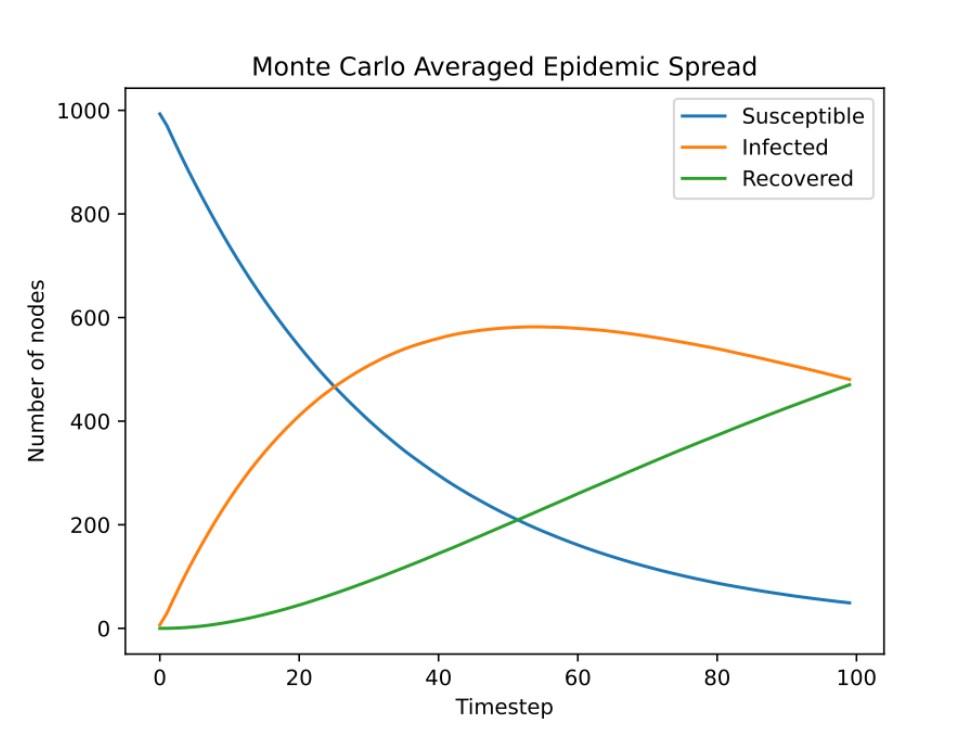

Figure 4: Monte Carlo Simulation (SIR)

Averaging removes random fluctuations and smoothens the curves, and as a result, it resembles the shape of a deterministic SIR model:

- the infected population follows a smooth rise, peak and decline pattern
- the susceptible population decreases steadily
- The recovered population increases slowly.

The averaged curves converge towards the expected epidemic trajectory for these parameters. This happens because the ER network acts like a well-mixed population. The expected trajectory is almost identical to the mean-field solutions one would get from deterministic models.

4.2 The Watts-Strogatz Network

4.2.1 SEIRD Implementation

To validate that the algorithm was working, the mean trajectories of 50 Monte Carlo runs were examined. Fig. 5 shows a graph of the number of nodes in each SEIRD state over a time of 100 days. The solid line represents the mean average of nodes per disease state, and the shaded band represents +/- one standard deviation from the mean. At the beginning there is little variation from the mean because every node is in the S phase except for a few that have been set to E. The nodes in E can make attempts to infect other nodes but because only a small number are in E, only a small number of infection attempts can be made, meaning the randomness is kept low. As the number of nodes in E and I increase, more infection attempts can be made. Nodes in I also have three options: remain infectious, recover, or die. A larger number of nodes at play coupled with multiple pathways for nodes compounds the randomness of the system, resulting in higher standard deviations. This shows the SERID implementation is working correctly, but from this graph alone we cannot determine the impact of the Watts-Strogatz network.

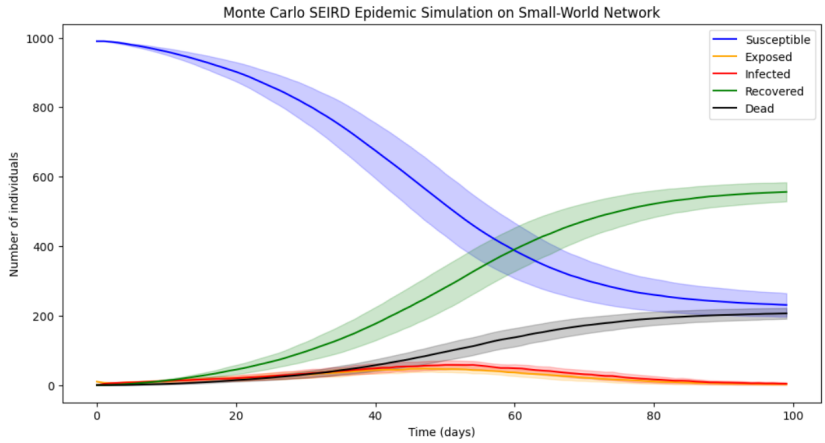

Figure 5: SEIRD implementation

4.2.2 Watts-Strogatz Implementation

To check if the Watts-Strogatz network is working we check two characteristics of the network: moderate clustering and short average path lengths. To do so we must have some benchmark to compare these parameters to. Watts-Strogatz sits somewhere between a regular lattice and a completely random graph (the Erdős-Rényi network). Regular lattices have very long path lengths because there are no shortcuts connecting nodes across the network, while random graphs have very short path lengths because they consist entirely of shortcuts. Regular lattices have high amounts of clustering as every node is connected to its k nearest neighbors, while random graphs have very low amounts of clustering. By generating a Watts-Strogatz network, a regular lattice network, and a random graph, we can confirm our Watts-Strogatz parameters are what we would expect. These results are shown in Fig 6. The clustering coefficient for the Watts-Strogatz network is 0.222, much higher than that of the random graph (clustering coefficient = 0.010) and considerably lower than that of the regular lattice (clustering coefficient = 0.667). The average path length between two nodes in the Watts-Strogatz network was 3.58, higher than the random graph (average path length = 3.25) and much lower than the random graph (average path length = 50.45). The path lengths for the Watts-Strogatz and random graph here are similar because there was a high rewiring probability in the Watts-Strogatz network (= 0.3).

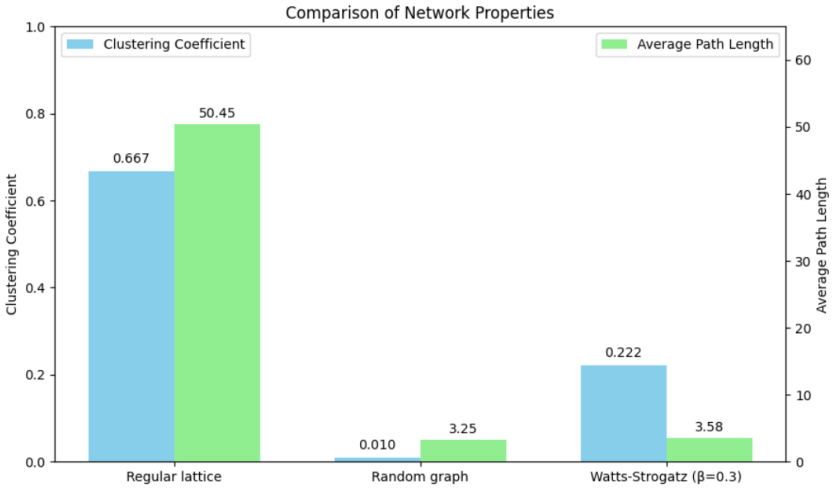

Figure 6: Watts-Strogatz Implementation

4.2.3 Random Walks using the Watts-Strogatz Network

To further investigate the dynamics of node-to-node interaction within the Watts-Strogatz network, a random walk model was used. A random walk simulates a stochastic process in which a "walker" moves stepwise from node to node by randomly selecting one of the neighbors at each time step. In the context of epidemic spreading, this can represent the movement of a pathogen through a population or the random contacts between individuals.

Simulations of 200 independent walkers were performed, each "walking" 300 steps on the Watts-Strogatz network. The network itself retained the parameters defined previously (N, k, β). For each walker, distance from the starting node at each step, computed using pre-calculated shortest path lengths was computed. The mean distance is the average distance on the shortest-path between all walkers at each step, while the mean squared displacement (MSD) emphasizes larger deviations, providing a measure of the variance of the spread.

Fig. 7 shows the mean distance and $\sqrt{MSD}$ over the course of 300 steps. Initially, both metrics grow linearly, as there are no constraints on where they can walk. Over time, growth begins to saturate, as each node can only be visited once. The small-world structure of the Watts-Strogatz network accelerates the diffusion relative to a regular lattice because of the presence of shortcuts, while moderate clustering slows the spread relative to a fully random network. These results demonstrate the combined effect of local clustering and network shortcuts on diffusion dynamics.

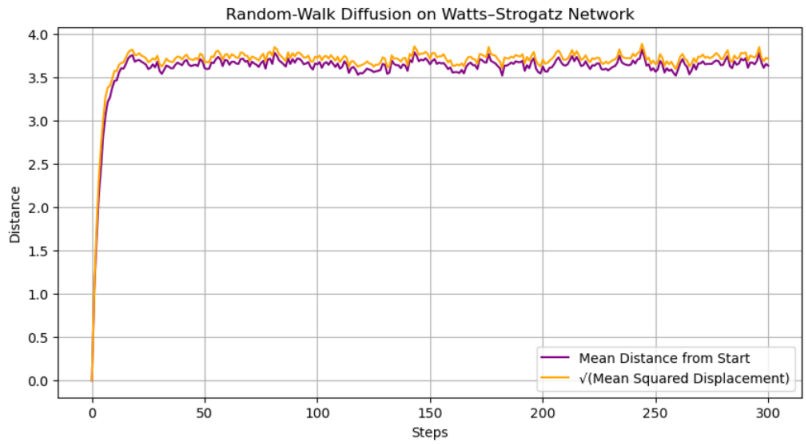

Figure 7: Random Walk Diffusion on Watts-Strogatz Network

4.3 Barabási–Albert Network

4.3.1 Network Graph

We implemented and plotted a graph of a Barabási–Albert network with N=100 nodes and set parameter m = 2. Here we can see some hubs and superspreader nodes in the middle of the graph, while most nodes are more sparsely connected. When adjusting m=4, we see a much more densely connected graph. M = 2 was kept for the rest of the investigation.

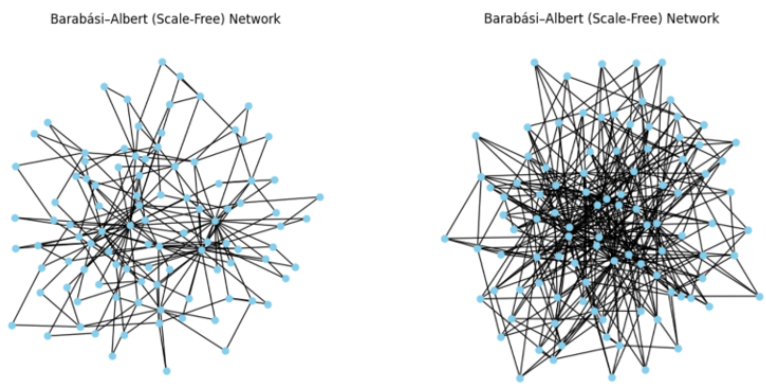

Figure 8: Barabási–Albert Graph (m=(2,4))

4.3.2 Degree Distribution

The degree distribution was plotted for 20 BA simulations on a log-log scale. While there's greater variability at the higher degrees, the degree distributions conform to a relatively-straight downward slope on a log-log scale, demonstrating that this scale-free degree distribution follows a power-law form, with a small number of high degree hubs.

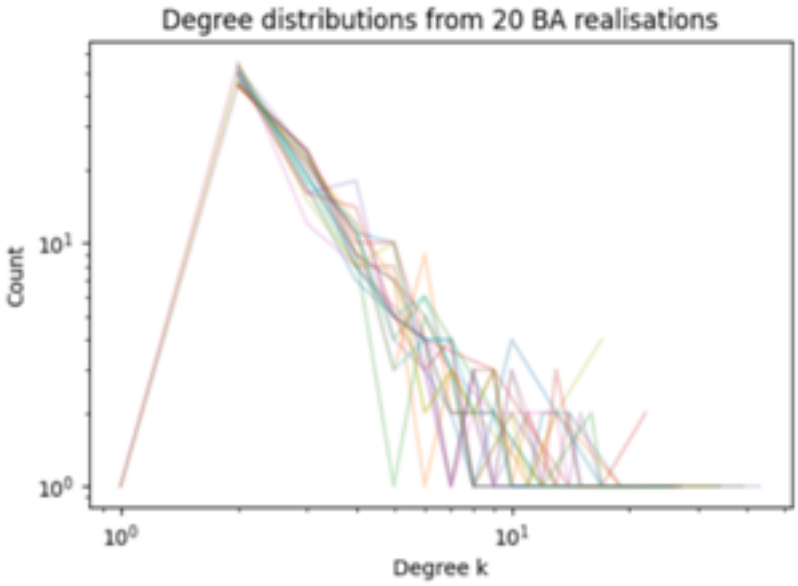

Figure 9: 20 Degree Distributions from BA Network

4.3.3 Single Random Walk (SIR implementation)

Two individual random walks through the network were performed with $\beta = 0.05$ and $\gamma = 0.02$. We see quite different peaks of infection, with the first at around 70 infected nodes, and the second just shy of 50. The BA network produces this variability, as the prevalence of an infection will be greatly affected by its success in infiltrating hubs.

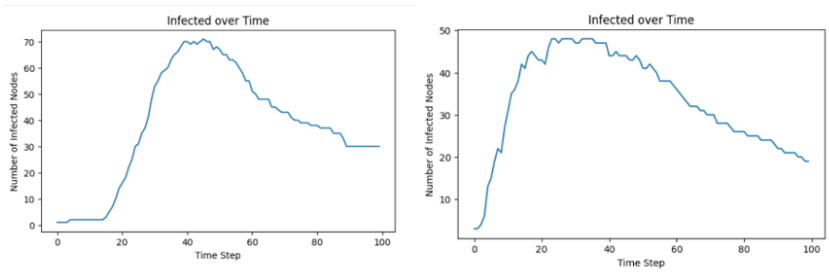

Figure 10: Two Simulations of a Random Walk of BA Network

4.3.4 Parameter Variation

The effect of varying β and γ values was investigated for the BA network. Monte Carlo Simulations of SIR processes were ran for different values of β and γ. Greater beta values are shown to increase peak prevalence, as well as total infection, and reduce the time take to reach its peak. Longer infections (lower recovery rate (γ)) also amplified and prolonged the epidemic.

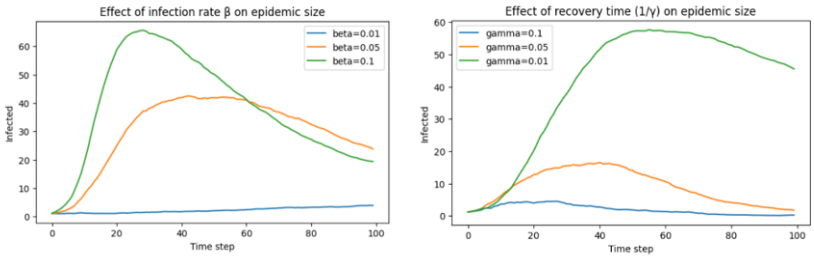

Figure 11: Monte Carlo SIR with varying β and γ

4.3.5 SEIRD Implementation

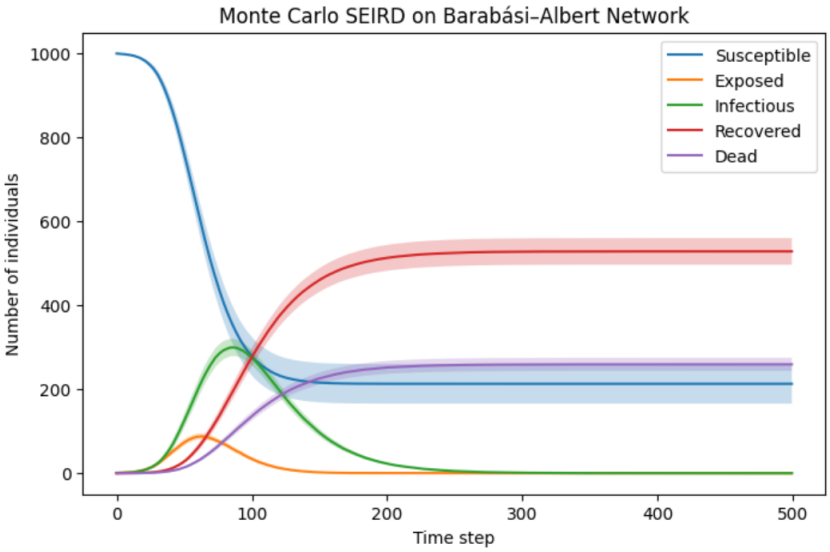

Figure 12: Monte Carlo SEIRD

Figure 11 shows the mean SEIRD epidemic trajectory on a Barabási–Albert scale-free network, together with 95% confidence intervals obtained from 200 Monte Carlo realisations. The susceptible population decreases rapidly while the infectious population grows. The presence of highly-connected hubs accelerates this initial growth. Once a hub becomes infected, it quickly propagates through its large number of contacts. The exposed population peaks early, without a very large peak, reflecting its nature as a relatively short transient state. It precedes the peak of the infectious state coming soon after. The infectious populations exhibits a pronounced early peak, characteristic of epidemic spreading in scale-free networks. The well-connected hubs create superspreader pathways through the network, leading to faster growth and a more concentrated infectious peak than in Erdős-Rényi or Watts-Strogatz networks. The dead and recovered curves rise sharply and plateau after the infectious curve has peaked and is returning near zero. These curves follow the same shape and pattern, with a lower magnitude for the dead curve relative to the recovered curve. Soon after 200 time steps, the environment is stable and the epidemic has passed. These results highlight the proliferation typical of this scale-free network, with rapid growth and large, early peaks.

5 Comparison to Related Algorithms

The models used above provide a clear demonstration of network-based stochastic simulations of epidemic spreading. However, there are many other approaches that are different in terms of structure, assumptions and level of complexity. This section compares and contrasts related algorithms that were not implemented in this project.

5.1 Deterministic Differential-Equation Models

Deterministic SIR and SEIR models (proposed by Kermack and McKendrick) [1] involve the use of continuous differential equations (described in Section 3) while relying on the assumption of homogenous mixing. Once these equations are solved, smooth epidemic curves as well as identical outcomes are produced, provided that the same parameters are used. This happens because there is no randomness, and the progression of the model is fully dependent on the differential equation.

In contrast, the models implemented in this project are discrete-time and stochastic in nature. Instead of solving differential equations, it simulates transmission and recovery as probabilistic events that happen on the edges of a network. As a result of this, each run shows randomness that deterministic models cannot catch. Our models can reveal outbreaks dying early by chance, superspreading and, among other things, structural influence from the underlying network model (e.g hubs). These things lie outside the reach of deterministic models, but are extremely important in understanding how real epidemics happen.

5.2 Continuous-Time Stochastic Models

Continuous-time stochastic models, such as the Gillespie method [2], simulate epidemic spreading by generating events based on each event's reaction rate. In contrast to the discrete-time models, which update the status of each individual at each timestep, the Gillespie method considers infection and recovery as Poisson processes and computes the exact time at which the next event (infection/recovery) will happen.

In the discrete-time, stochastic models, the occurrence of infection or recovery events is determined based on probability at each timestep. While this discrete-time approach is simpler and computationally cheaper than the Gillespie algorithm, it does not pick up exact event timings and can introduce timestep-dependent artefacts. For example, in the ER model above, let's say one time step equals one day. This means that only at the end of that day will someone become infected or recover, or nothing happens. But in real life, this doesn't happen, as events can happen throughout the whole day. If deciding between the two models, model choice would depend on the experimental goal. For greater accuracy, continuous-time models are better suited. However, discrete-time models are faster, simpler and more computationally efficient.

5.3 Analytical network models

There are also analytical approaches to studying epidemics using tools such as probability theory and mean-field approximation. An example of this is the degree-based mean-field model, which involves the grouping of nodes based on their degree, and deriving equations for infection and recovery rates within each degree group.

These methods can provide expressions for values of importance, such as epidemic thresholds and final outbreak size. As an example, a group of researchers [3] showed that the spread of disease on random networks can be analysed using percolation theory, allowing the epidemic

threshold to be expressed in terms of the first and second moments of the degree distribution. Another group of researchers [4] showed scale-free networks with heavy-tailed degree distributions (not clustered) may have near non-existent epidemic thresholds, meaning that even infections with very small transmission probabilities can sustain large outbreaks.

The models used in the project rely on explicit stochastic simulations of disease spread on finite networks. While analytical models provide concise mathematical results and several insights, they generally rely on assumptions such as large and infinite networks, the absence of clustering, and uniform behaviour among nodes with the same degree group. They also do not capture run-to-run variability, random epidemic die out, or structural effects such as local clustering. Monte Carlo simulations would complement analytical network models well by revealing the stochastic behaviour of epidemics on finite networks.

6 Use of Monte Carlo Simulations in Epidemic Spreading Research

In this report we have described how monte carlo simulations are used to model epidemics. With regards to why, monte carlo simulations are used to model different scenarios that could occur in an epidemic, and how outcomes change when certain parameters are introduced/adjusted. For example, a popular use of MCS is to track how different vaccine administration programmes can alter disease spread. In a study examining the impact of vaccines on Covid-19 outcomes, different vaccines (Moderna, Pfizer, and Janssen) were included in the simulation to see which vaccine would result in the lowest total number of infections [1]. They also investigated how factors such as vaccine rates and efficacy could impact epidemic peaks and total number of infections.

MCS can also be used to see which diagnostic tests are most effective at detecting disease cases. This strategy was used in a 2020 study comparing the cost and efficacy of three malaria diagnostic strategies (microscopy, rapid diagnostic tests (RDTs), microscopy and RDTs)[2]. They found that RDTs were the most effective but also the most costly. This type of information can be very valuable when designing diagnostic policies.

The two above examples involved adding a different parameter representing vaccination/diagnostics, but another way MCS are used in research is by adjusting one of the standard parameters. In our code demonstrations, we set parameters like beta (transmission rate), sigma (incubation time), etc as constants. However, real epidemics rarely have a fixed transmission rate, and researchers often use Monte Carlo methods to estimate how transmission changes over time. For example, in a recent study analysing Ebola case data, a stochastic SIR model with a time-varying transmission rate was fitted using a Markov chain Monte Carlo algorithm. Instead of setting beta to different values manually, the algorithm inferred when transmission increased or decreased during the outbreak. The results showed clear shifts in transmission that aligned with real-world interventions (such as school closures and border restrictions), demonstrating how MCS can uncover the timing and magnitude of behavioural or policy-driven changes in disease spread.

7 Conclusion

This project investigated how random walks spread over different classes of networks, with a particular focus on modelling epidemic spreading. Erdős-Rényi, Watts-Strogatz and Barabási–Albert models were compared to understand how network structure influenced the speed and scale of epidemic diffusion. Monte Carlo simulations were key to producing average-case scenarios for each network and dealing with the randomness of any given iteration. The Erdős-Rényi network introduced us to modelling epidemic curves across a homogenous population. Vary-

ing β showed, predictably, that larger values produced greater peaks for the outbreaks, and also caused this peak to arrive sooner. The stochasticity of individual runs was handled with Monte Carlo simulations producing smooth, stable mean trajectories. The Watts-Strogatz network was investigated and validated by analysing its clustering coefficient and average path length. Its moderate clustering and short but non-minimal path lengths positioned it between a regular lattice and a completely random Erdős-Rényi graph, and exhibited its small-world properties. This local clustering slowed early spread, but the presence of short cuts in the graph still generated a considerable death toll, once the infection escaped its initial cluster. The Barabási–Albert model showed the strongest departure from homogenous behaviour. Plotting the degree distribution on a log-log scale revealed the expected power-law form, where a few hubs of high degree exist, and most nodes are of lower degree. The hubs play a key role in infection spreading, as diseases that enter these hubs grow rapidly and infect a large portion of the network. Higher infection probabilities β and slower recovery rates γ amplified growth by allowing hubs to infect more neighbours before recovery. Our findings highlight that network structure is a dominant factor in epidemic behaviour. Erdős-Rényi networks represent a homogenously mixed population, Watts-Strogatz networks balance local clustering with short global distances, and Barabási–Albert networks produce hubs where a population is vulnerable to superspreading. By simulating many outbreaks with these networks, we've shown their structural differences, and that adjusting their parameters can greatly affect the outbreak's trajectory. This demonstrates the importance of network-based modelling in modern epidemiology.

8 References

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9 Appendix

Please see Jupyter notebook for code